6CS012 - Worksheet 4: Exercise

Perfect Bansal

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

Task 1: Data Preparation

Loading the Devnagari digits dataset using PIL, converting images to NumPy arrays, normalizing to [0, 1], and extracting labels from folder names.

```
# Import required libraries
import os
import numpy as np
import tensorflow as tf
from tensorflow.keras.utils import to_categorical
import matplotlib.pyplot as plt
from PIL import Image

print(f"TensorFlow version: {tf.__version__}")
print(f"Keras version: {tf.keras.__version__}")
```

```
TensorFlow version: 2.19.0
Keras version: 3.13.2
```

```
train_dir = "/content/drive/MyDrive/AI Sem6/DevanagariHandwrittenDigitDataset/Train"
test_dir = "/content/drive/MyDrive/AI Sem6/DevanagariHandwrittenDigitDataset/Test"

# Define image size
img_height, img_width = 28, 28
```

```
# Function to load images and labels using PIL
def load_images_from_folder(folder):
    """
    Loads images from a folder structure where each subfolder represents a class.

    Parameters:
        folder (str): Path to the root folder containing class subfolders.

    Returns:
        images (np.array): Array of normalized image data.
        labels (np.array): Array of integer labels.
    """
    images = []
    labels = []

    # Get sorted class names (digit_0, digit_1, ..., digit_9)
    class_names = sorted(os.listdir(folder))
    # Map class folder names to integer labels
    class_map = {name: i for i, name in enumerate(class_names)}

    print(f"Found classes: {class_names}")
    print(f"Class mapping: {class_map}")

    for class_name in class_names:
        class_path = os.path.join(folder, class_name)
        if not os.path.isdir(class_path):
            continue

        label = class_map[class_name]
        file_count = 0
```

```
        for filename in os.listdir(class_path):
            img_path = os.path.join(class_path, filename)
            try:
                # Load image using PIL
                img = Image.open(img_path).convert("L")  # Convert to grayscale
                img = img.resize((img_width, img_height))  # Resize to 28x28
                img = np.array(img) / 255.0  # Normalize pixel values to [0, 1]
                images.append(img)
                labels.append(label)
                file_count += 1
            except Exception as e:
                print(f"Error loading {img_path}: {e}")

        print(f"  Loaded {file_count} images for class '{class_name}' (label={label})")

    return np.array(images), np.array(labels)
```

```
# Load training and testing datasets
print("Loading training data...")
x_train, y_train = load_images_from_folder(train_dir)
print(f"\nLoading testing data...")
x_test, y_test = load_images_from_folder(test_dir)

print(f"\n--- Dataset Summary ---")
print(f"Training samples: {x_train.shape[0]}")
print(f"Testing samples: {x_test.shape[0]}")
print(f"Image shape: {x_train[0].shape}")
print(f"Label range: {y_train.min()} to {y_train.max()}")
```

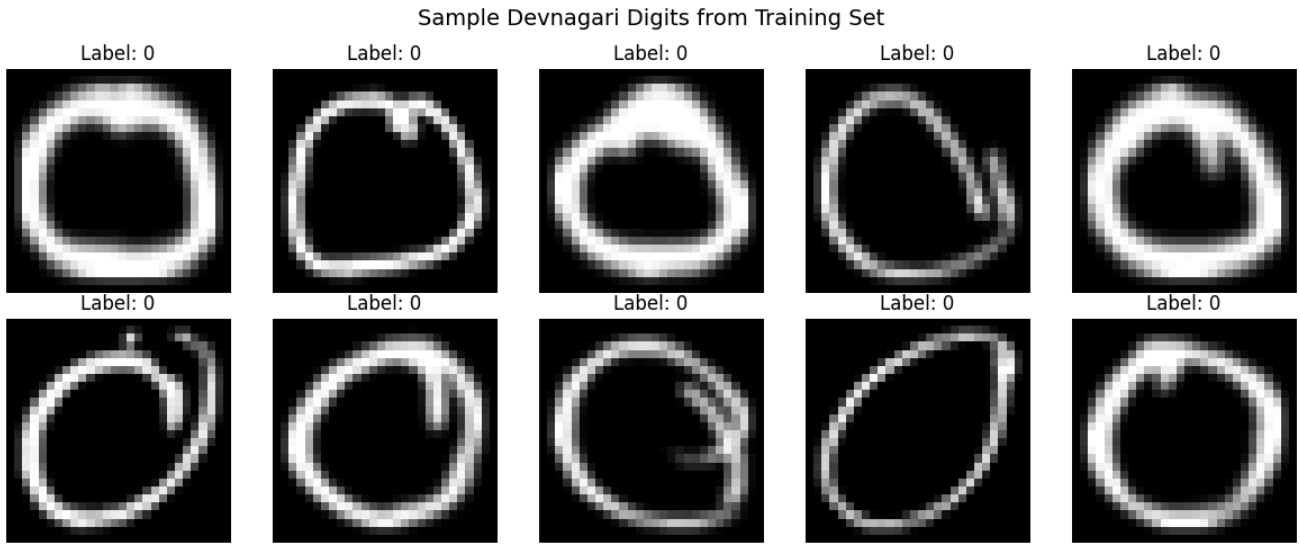
```
Loading training data...
Found classes: ['digit_0', 'digit_1', 'digit_2', 'digit_3', 'digit_4', 'digit_5', 'digit_6', 'digit_7', 'digit_8', 'digit_9']
Class mapping: {'digit_0': 0, 'digit_1': 1, 'digit_2': 2, 'digit_3': 3, 'digit_4': 4, 'digit_5': 5, 'digit_6': 6, 'digit_7': 7,
  Loaded 1700 images for class 'digit_0' (label=0)
  Loaded 1700 images for class 'digit_1' (label=1)
  Loaded 1700 images for class 'digit_2' (label=2)
  Loaded 1700 images for class 'digit_3' (label=3)
  Loaded 1700 images for class 'digit_4' (label=4)
  Loaded 1700 images for class 'digit_5' (label=5)
  Loaded 1700 images for class 'digit_6' (label=6)
  Loaded 1700 images for class 'digit_7' (label=7)
  Loaded 1700 images for class 'digit_8' (label=8)
  Loaded 1700 images for class 'digit_9' (label=9)

Loading testing data...
Found classes: ['digit_0', 'digit_1', 'digit_2', 'digit_3', 'digit_4', 'digit_5', 'digit_6', 'digit_7', 'digit_8', 'digit_9']
Class mapping: {'digit_0': 0, 'digit_1': 1, 'digit_2': 2, 'digit_3': 3, 'digit_4': 4, 'digit_5': 5, 'digit_6': 6, 'digit_7': 7,
  Loaded 300 images for class 'digit_0' (label=0)
  Loaded 300 images for class 'digit_1' (label=1)
  Loaded 300 images for class 'digit_2' (label=2)
  Loaded 300 images for class 'digit_3' (label=3)
  Loaded 300 images for class 'digit_4' (label=4)
  Loaded 300 images for class 'digit_5' (label=5)
  Loaded 300 images for class 'digit_6' (label=6)
  Loaded 300 images for class 'digit_7' (label=7)
  Loaded 300 images for class 'digit_8' (label=8)
  Loaded 300 images for class 'digit_9' (label=9)

--- Dataset Summary ---
Training samples: 17000
Testing samples: 3000
Image shape: (28, 28)
Label range: 0 to 9
```

```
# Flatten the 28x28 images into 784-dimensional vectors for FCN input
x_train_flat = x_train.reshape(-1, img_height * img_width)
x_test_flat = x_test.reshape(-1, img_height * img_width)

# One-hot encode labels (10 classes: digits 0-9)
y_train_encoded = to_categorical(y_train, num_classes=10)
y_test_encoded = to_categorical(y_test, num_classes=10)

print(f"Flattened training data shape: {x_train_flat.shape}")
print(f"Flattened testing data shape: {x_test_flat.shape}")
print(f"One-hot encoded training labels shape: {y_train_encoded.shape}")
```

```
print(f"One-hot encoded testing labels shape: {y_test_encoded.shape}")
```

```
Flattened training data shape: (17000, 784)
Flattened testing data shape: (3000, 784)
One-hot encoded training labels shape: (17000, 10)
One-hot encoded testing labels shape: (3000, 10)
```

```
# Visualize sample images from the training set
plt.figure(figsize=(12, 5))
for i in range(10):
    plt.subplot(2, 5, i + 1)
    plt.imshow(x_train[i], cmap='gray')
    plt.title(f"Label: {y_train[i]}")
    plt.axis("off")
plt.suptitle("Sample Devnagari Digits from Training Set", fontsize=14)
plt.tight_layout()
plt.show()
```



Task 2: Build the FCN Model

Creating a Sequential model with:

- Input: 784 features (flattened 28×28 images)
- Hidden Layer 1: 64 neurons, sigmoid activation
- Hidden Layer 2: 128 neurons, sigmoid activation
- Hidden Layer 3: 256 neurons, sigmoid activation
- Output Layer: 10 neurons, softmax activation

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Input

# Build the Sequential FCN model
model = Sequential([
    Input(shape=(784,)),                        # Input layer: 784 features
    Dense(64, activation="sigmoid"),            # Hidden layer 1: 64 neurons
    Dense(128, activation="sigmoid"),           # Hidden layer 2: 128 neurons
    Dense(256, activation="sigmoid"),           # Hidden layer 3: 256 neurons
    Dense(10, activation="softmax"),            # Output layer: 10 classes
])

# Display model architecture
model.summary()
```

```
Model: "sequential"
```

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 64)	50,240
dense_1 (Dense)	(None, 128)	8,320
dense_2 (Dense)	(None, 256)	33,024
dense_3 (Dense)	(None, 10)	2,570

```
Total params: 94,154 (367.79 KB)
Trainable params: 94,154 (367.79 KB)
Non-trainable params: 0 (0.00 B)
```

Task 3: Compile the Model

Using:

- **Optimizer:** Adam (adaptive learning rate, works well for most tasks)
- **Loss:** Categorical Crossentropy (multi-class classification with one-hot encoded labels)
- **Metric:** Accuracy

```
# Compile the model
model.compile(
    optimizer="adam",
    loss="categorical_crossentropy",
    metrics=["accuracy"]
)

print("Model compiled successfully.")
```

```
Model compiled successfully.
```

Double-click (or enter) to edit

```
from tensorflow.keras.callbacks import ModelCheckpoint, EarlyStopping

# Define callbacks
callbacks = [
    ModelCheckpoint(
        filepath="best_devnagari_model.keras",
        monitor="val_loss",
        save_best_only=True,
        verbose=1
    ),
    EarlyStopping(
        monitor="val_loss",
        patience=5,
        restore_best_weights=True,
        verbose=1
    ),
]

# Train the model
history = model.fit(
    x_train_flat,
    y_train_encoded,
    batch_size=128,
    epochs=20,
    validation_split=0.2,
    callbacks=callbacks,
    verbose=1
)
```

```
Epoch 1/20
101/107 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.2677 - loss: 2.0010
Epoch 1: val_loss improved from None to 7.11396, saving model to best_devnagari_model.keras

Epoch 1: finished saving model to best_devnagari_model.keras
107/107 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step - accuracy: 0.4648 - loss: 1.6341 - val_accuracy: 0.0000e+00 - val_loss: 7.1140
Epoch 2/20
```

```
 98/107 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8326 - loss: 0.6471
Epoch 2: val_loss did not improve from 7.11396
107/107 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - accuracy: 0.8689 - loss: 0.4967 - val_accuracy: 0.0000e+00 - val_loss: 8.7936
Epoch 3/20
100/107 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9270 - loss: 0.2725
Epoch 3: val_loss did not improve from 7.11396
107/107 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - accuracy: 0.9318 - loss: 0.2485 - val_accuracy: 0.0000e+00 - val_loss: 9.4268
Epoch 4/20
101/107 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9466 - loss: 0.1912
Epoch 4: val_loss did not improve from 7.11396
107/107 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - accuracy: 0.9512 - loss: 0.1762 - val_accuracy: 0.0000e+00 - val_loss: 9.9300
Epoch 5/20
 99/107 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9587 - loss: 0.1407
Epoch 5: val_loss did not improve from 7.11396
107/107 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - accuracy: 0.9596 - loss: 0.1395 - val_accuracy: 0.0000e+00 - val_loss: 10.3371
Epoch 6/20
107/107 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9675 - loss: 0.1131
Epoch 6: val_loss did not improve from 7.11396
107/107 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - accuracy: 0.9675 - loss: 0.1126 - val_accuracy: 0.0000e+00 - val_loss: 10.3610
Epoch 6: early stopping
Restoring model weights from the end of the best epoch: 1.
```

Visualizing Training Progress

```
# Plot training and validation loss & accuracy
train_loss = history.history['loss']
val_loss = history.history['val_loss']
train_acc = history.history['accuracy']
val_acc = history.history['val_accuracy']

epochs_range = range(1, len(train_loss) + 1)

plt.figure(figsize=(14, 5))

# Loss plot
plt.subplot(1, 2, 1)
plt.plot(epochs_range, train_loss, label='Training Loss', color='blue', linewidth=2)
plt.plot(epochs_range, val_loss, label='Validation Loss', color='orange', linewidth=2)
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.title('Training and Validation Loss')
plt.legend()
plt.grid(True, alpha=0.3)

# Accuracy plot
plt.subplot(1, 2, 2)
plt.plot(epochs_range, train_acc, label='Training Accuracy', color='blue', linewidth=2)
plt.plot(epochs_range, val_acc, label='Validation Accuracy', color='orange', linewidth=2)
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.title('Training and Validation Accuracy')
plt.legend()
plt.grid(True, alpha=0.3)

plt.suptitle('Model Training Progress', fontsize=14, y=1.02)
plt.tight_layout()
plt.show()
```

Evaluating on the test set to measure generalization performance.

```
# Evaluate the model on test data
test_loss, test_acc = model.evaluate(x_test_flat, y_test_encoded, verbose=2)

print(f"\n--- Test Results ---")
print(f"Test Loss: {test_loss:.4f}")
print(f"Test Accuracy: {test_acc:.4f}")
```

```
94/94 - 0s - 2ms/step - accuracy: 0.6457 - loss: 2.0887

--- Test Results ---
Test Loss: 2.0887
Test Accuracy: 0.6457
```

Saving the trained model to an `.h5` file and reloading it to verify.

```
# Save the model in H5 format
model.save('devnagari_fcn_model.h5')
print("Model saved as 'devnagari_fcn_model.h5'")

# Load the saved model
loaded_model = tf.keras.models.load_model('devnagari_fcn_model.h5')
print("Model loaded successfully.")

# Re-evaluate the loaded model to confirm it works correctly
loaded_loss, loaded_acc = loaded_model.evaluate(x_test_flat, y_test_encoded, verbose=2)
print(f"\n--- Loaded Model Test Results ---")
print(f"Test Loss: {loaded_loss:.4f}")
print(f"Test Accuracy: {loaded_acc:.4f}")
```

```
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format
WARNING:absl:Compiled the loaded model, but the compiled metrics have yet to be built. `model.compile_metrics` will be empty unt
Model saved as 'devnagari_fcn_model.h5'
Model loaded successfully.
94/94 - 1s - 6ms/step - accuracy: 0.6457 - loss: 2.0887

--- Loaded Model Test Results ---
Test Loss: 2.0887
Test Accuracy: 0.6457
```

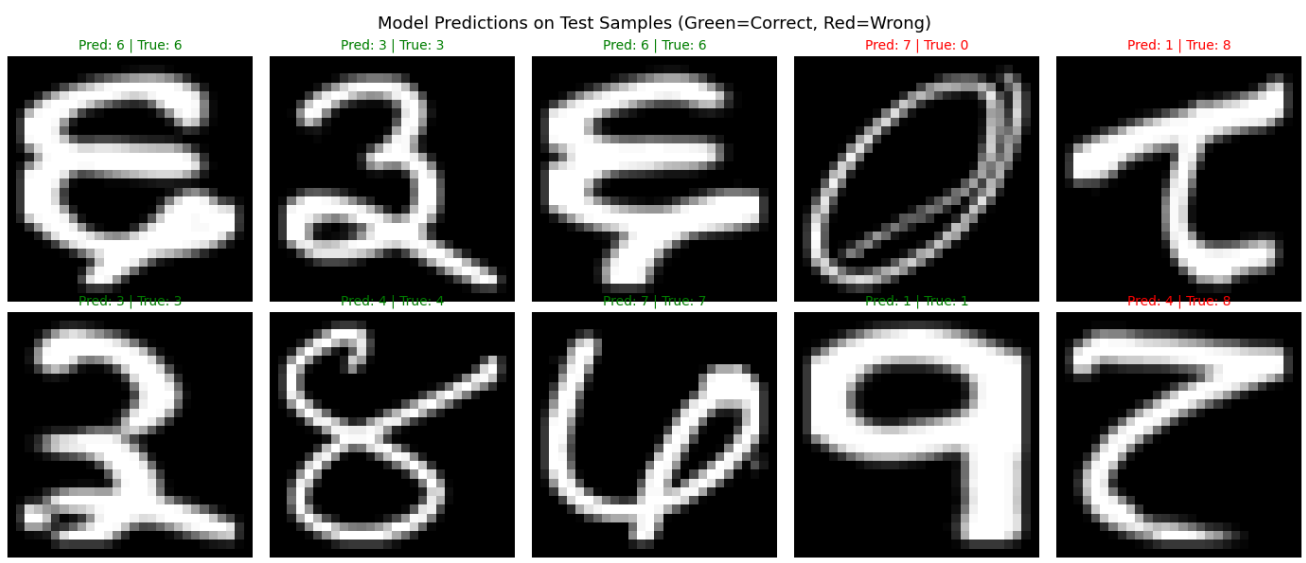
Double-click (or enter) to edit

```
# Make predictions on test data
predictions = model.predict(x_test_flat)

# Convert predicted probabilities to digit labels
```

```
predicted_labels = np.argmax(predictions, axis=1)
true_labels = np.argmax(y_test_encoded, axis=1)

# Display first 5 predictions
print("--- Sample Predictions ---")
for i in range(5):
    print(f"Image {i+1}: Predicted = {predicted_labels[i]}, True = {true_labels[i]}, "
          f"{'✓ Correct' if predicted_labels[i] == true_labels[i] else '✗ Wrong'}")
```

```
94/94 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step
--- Sample Predictions ---
Image 1: Predicted = 0, True = 0, ✓ Correct
Image 2: Predicted = 0, True = 0, ✓ Correct
Image 3: Predicted = 0, True = 0, ✓ Correct
Image 4: Predicted = 0, True = 0, ✓ Correct
Image 5: Predicted = 0, True = 0, ✓ Correct
```

```
# Visualize predictions on sample test images
plt.figure(figsize=(14, 6))
# Pick 10 random test samples
np.random.seed(42)
sample_indices = np.random.choice(len(x_test), 10, replace=False)

for i, idx in enumerate(sample_indices):
    plt.subplot(2, 5, i + 1)
    plt.imshow(x_test[idx], cmap='gray')

    pred = predicted_labels[idx]
    true = true_labels[idx]
    color = 'green' if pred == true else 'red'

    plt.title(f"Pred: {pred} | True: {true}", color=color, fontsize=10)
    plt.axis("off")

plt.suptitle("Model Predictions on Test Samples (Green=Correct, Red=Wrong)", fontsize=13)
plt.tight_layout()
plt.show()
```




```
# Print overall prediction statistics
correct = np.sum(predicted_labels == true_labels)
total = len(true_labels)
print(f"\n--- Prediction Summary ---")
print(f"Correct predictions: {correct}/{total}")
print(f"Overall accuracy: {correct/total:.4f}")

# Per-class accuracy
print(f"\n--- Per-Class Accuracy ---")
```

```
for digit in range(10):
    mask = true_labels == digit
    class_correct = np.sum(predicted_labels[mask] == true_labels[mask])
    class_total = np.sum(mask)
    if class_total > 0:
        print(f"Digit {digit}: {class_correct}/{class_total} = {class_correct/class_total:.4f}")
```

```
--- Prediction Summary ---
Correct predictions: 1937/3000
Overall accuracy: 0.6457

--- Per-Class Accuracy ---
Digit 0: 253/300 = 0.8433
Digit 1: 294/300 = 0.9800
Digit 2: 230/300 = 0.7667
Digit 3: 144/300 = 0.4800
Digit 4: 285/300 = 0.9500
Digit 5: 210/300 = 0.7000
Digit 6: 242/300 = 0.8067
Digit 7: 279/300 = 0.9300
Digit 8: 0/300 = 0.0000
Digit 9: 0/300 = 0.0000
```

Summary

This notebook completed all 7 tasks of the Worksheet 4 exercise:

1. **Data Preparation** — Loaded Devnagari digit images using PIL, normalized to [0,1], and one-hot encoded labels.
2. **Model Building** — Created a Sequential FCN with 3 hidden layers (64→128→256 neurons, sigmoid activation) and a softmax output layer.
3. **Compilation** — Compiled with Adam optimizer, categorical crossentropy loss, and accuracy metric.
4. **Training** — Trained for 20 epochs with batch size 128, 20% validation split, and ModelCheckpoint/EarlyStopping callbacks.
5. **Evaluation** — Evaluated test accuracy and loss using `model.evaluate()`.
6. **Save/Load** — Saved to `.h5` file and reloaded to verify performance.
7. **Predictions** — Generated predictions using `model.predict()`, visualized results, and computed per-class accuracy.